Units 3 and 4

Practice exam

Student name: _____

Structure of this paper

Section	Number of questions	Approximate time to allow for answering this section	Marks
A Multiple choice	30	40 minutes	30
B Short answer	11	110 minutes	90
	Total	**2.5 hours**	**120 marks**

Materials supplied

- Data book
- Answer sheet for multiple-choice questions

Instructions

- You will have the following time for the examination:
  - reading time—15 minutes
  - writing time—2.5 hours.
- Write your name in the space provided above.
- The following items are approved for use in the examination:
  - pens (black preferred) for writing, pencils (at least 2B) for drawing diagrams, highlighters, erasers, sharpeners, rulers and one scientific calculator.
- The following items are NOT permitted for use in the examination:
  - blank sheet of paper and/or correction fluid/tape.
- Mobile phones and/or any other unauthorised electronic devices are not permitted in the examination.
- Answer all multiple-choice questions using the answer sheet provided.
- Answer all short-answer questions in the spaces provided.
- Diagrams are not drawn to scale (unless otherwise stated).
- All written responses must be in English.

Disclaimer

This is a practice examination. It represents Pearson Australia's view only of what would be useful preparation material for the externally assessed examination.

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Section A—Multiple-choice questions

30 questions

30 marks

Allow about 40 minutes for this section.

Instructions for Section A

Answer all questions in pencil on the answer sheet provided for multiple-choice questions.

Choose the response that is **correct** or that **best answers** the question.

A correct answer scores 1; an incorrect response scores 0.

Marks will **not** be deducted for incorrect answers.

No marks will be given if more than one answer is completed for any question.

Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Question 1

Which of the following reactions will have the highest atom economy?

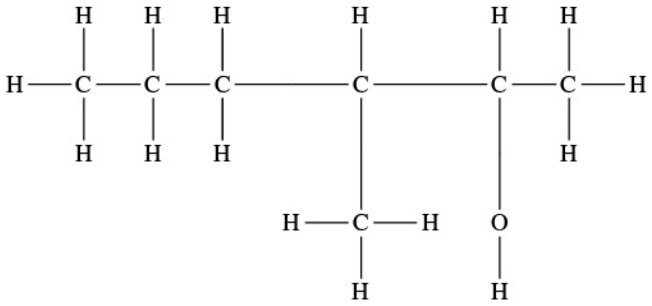
A. substitution of chlorine with propane

B. reaction between propene and bromine

C. production of methyl propanoate from methanol and propanoic acid

D. production of propan-1-ol by substitution of 1-chloropropane

Question 2



The systematic name for the molecule shown is:

A. 4-methylhexan-5-ol.

B. 3-methylhexan-2-ol.

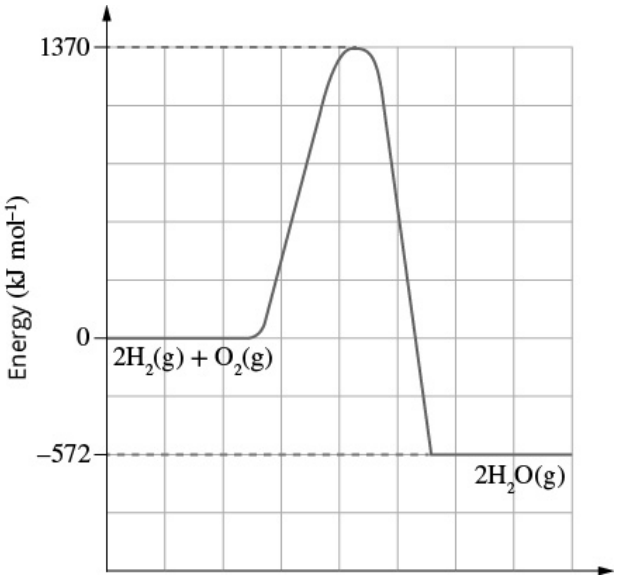
C. heptan-3-ol.

D. 1,3-dimethylpentan-1-ol.

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 3

The energy profile diagram for a chemical reaction is shown below.



What is the activation energy of the reverse reaction, in kJ mol^{-1}?

A. –1370

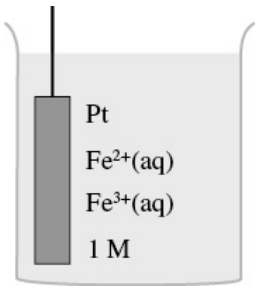
B. –572

C. +1370

D. +1942

Question 4

The diagram below shows a half-cell.



A half-equation that could occur in this cell is:

A. $Pt(s) \rightarrow Pt^{2+}(aq) + 2e^-$

B. $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$

C. $Fe^{3+}(aq) \rightarrow Fe^{2+}(aq) + e^-$

D. $Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + e^-$

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 5

New medicines being used to treat diseases, including Alzheimer's, are called competitive enzyme inhibitors. They work by:

A. changing the shape of the active site.

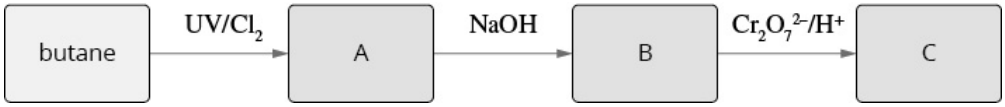
B. creating an alternative active site in the enzyme.

C. denaturing the enzyme.

D. binding to the enzyme active site to block another compound from doing so.

Question 6

A chemical flow chart that starts with the molecule butane is shown below.



Molecule C, formed at the end of this flow chart, will be:

A. 1-chlorobutane.

B. butan-1-ol.

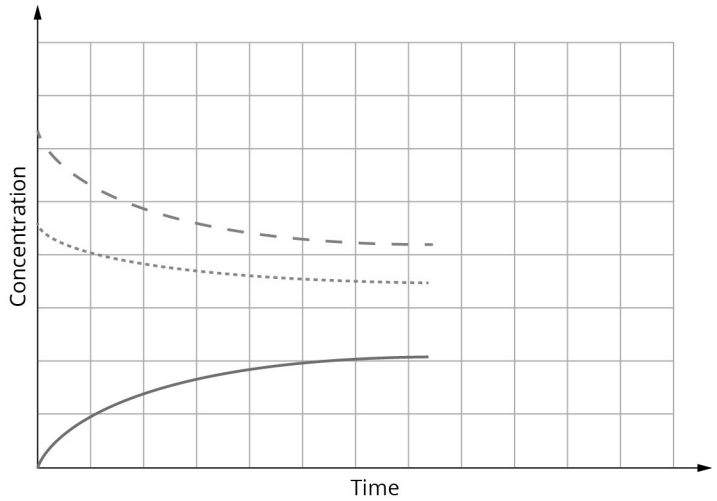
C. butanoic acid.

D. 1,2-dichlorobutane.

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 7

The concentrations of species as a reaction reaches equilibrium are shown in the graph below.



Using the accurate scale of the graph, select the reaction with the correct stoichiometric ratios of reactants and products to match the changes in concentration shown.

A. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

B. $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

C. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

D. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Question 8

Nitrosyl chloride can decompose to nitrogen monoxide and chlorine gas. The equation for the reaction is:

$2NOCl(g) \rightleftharpoons 2NO(g) + Cl_2(g)$

1.6 mol of NOCl is added to an empty reaction vessel. When equilibrium is reached, the amount of NOCl has dropped to 1.0 mol. The total number of moles of the substances in the reaction vessel at equilibrium will be:

A. 1.9 mol.

B. 2.2 mol.

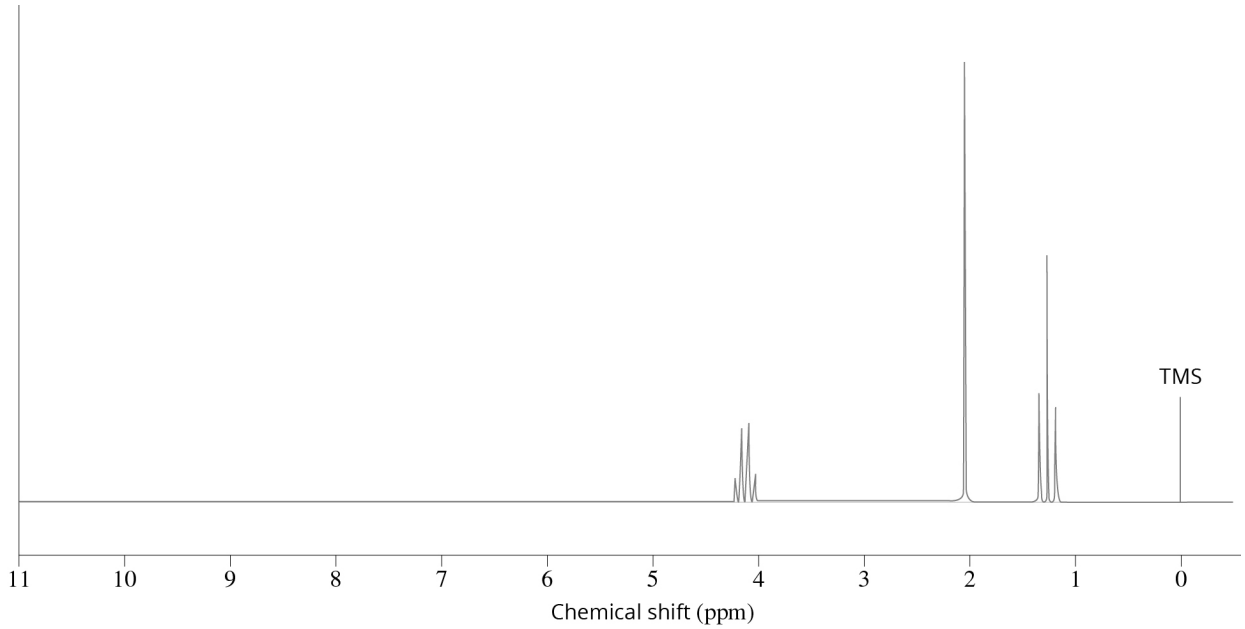
C. 4.8 mol.

D. 5.0 mol.

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 9

A proton (^{1}H) NMR is shown below.



This NMR could be for:

A. propanol.

B. methyl ethanoate.

C. ethyl ethanoate.

D. butanoic acid.

Question 10

The mass spectra of propane and carbon dioxide are compared. The spectra:

A. will be identical as both molecules have the same molar mass.

B. will have no peaks in common as they are very different molecules.

C. might have a strong peak at 29 for both molecules due to the fragmentation patterns.

D. might share a similar parent molecular ion but differ in most other peaks.

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 11

The heat of combustion of ethanol (C_2H_5OH) (M = 46.0 g mol^{-1}) is –1360 kJ mol^{-1}. The mass of ethanol required to raise the temperature of 120 g of water by 10.0°C will be:

A. 0.003 69 g

B. 0.170 g

C. 1.70 g

D. 120 g

Question 12

Chemical reactions involve the breaking of bonds and the formation of new bonds. Consider the formation of ammonia from nitrogen and hydrogen gases:

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

In this exothermic process:

A. the bond changes are all exothermic as the reaction itself is exothermic.

B. the breaking of bonds in nitrogen and hydrogen is endothermic and the formation of N–H bonds exothermic.

C. the breaking of bonds in nitrogen and hydrogen is exothermic and the formation of N–H bonds endothermic.

D. the overall reaction is exothermic but all changes occurring in bonds will be endothermic.

Question 13

Deduce the oxidation state of vanadium in NH_4VO_3.

A. +2

B. +3

C. +4

D. +5

Question 14

Which of the following statements about primary cells and fuel cells is correct?

A. Oxidation occurs at the anode and cations move into the cell.

B. Oxidation occurs at the anode and anions move into the cell.

C. Reduction occurs at the cathode and anions move into the cell.

D. Reduction occurs at the anode and cations move into the cell.

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 15

Lipids and carbohydrates provide energy to the body when metabolised. What is the name given to the reaction responsible for breaking down these complex structures into smaller molecules?

A. condensation

B. hydrolysis

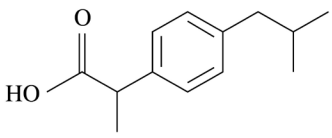
C. addition

D. oxidation

Question 16

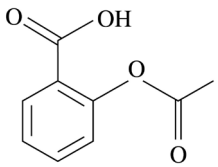
The four compounds below are found in common pain relief medicines. Which of these compounds contains 2 chiral centres?

A.



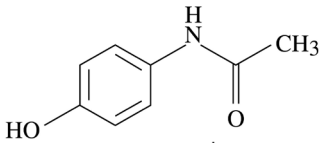
ibuprofen

B.



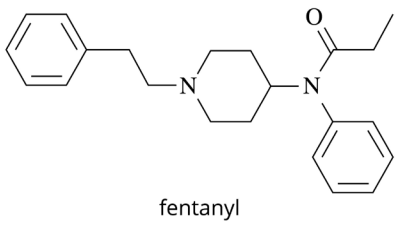
aspirin

C.



paracetamol

D.



fentanyl

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 17

Identify the protein structure that is the overall three-dimensional shape of an enzyme.

A. peptide link

B. primary

C. secondary

D. tertiary

Question 18

The type of chemical reaction by which biodiesel is produced from fats or oils is called:

A. transesterification.

B. combustion.

C. fermentation.

D. polymerisation.

Question 19

The rechargeable nickel–cadmium cell is used to power small appliances such as portable computers. When the cell is being used, the electrode reactions are represented by the following equations.

$NiO_2(s) + 2H_2O(l) + 2e^- \rightarrow Ni(OH)_2(s) + 2OH^-(aq)$

$Cd(s) + 2OH^-(aq) \rightarrow Cd(OH)_2(s) + 2e^-$

Which of the following occurs during the recharging of the nickel–cadmium cell?

I Cadmium is deposited on the negative electrode.

II The pH of the electrolyte increases .

III The direction of electron flow in the external circuit is from the anode to the cathode.

A. I only

B. I and II only

C. II and III only

D. I and III only

Question 20

Which of the following molecules is the most volatile?

A. hexan-2-one

B. 2-chlorohexane

C. 2-methylpentanal

D. propyl propanoate

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 21

Which one of the following classes of electrochemical cells involves only a non-spontaneous redox reaction?

A. fuel cells

B. electrolytic cells

C. primary galvanic cells

D. secondary galvanic cells

Question 22

In response to a pain stimulus, the brain produces small polypeptide molecules. These molecules block the transmission of pain through the central nervous system. The amino acid sequence in one such molecule is:

Met–Ala–Asp–Lys–Ile

The number of amino, carboxyl and amide functional groups within this sequence of the polypeptide is:

	Amino	**Carboxyl**	**Amide**
A.	1	1	5
B.	2	2	5
C.	1	1	4
D.	2	2	4

Question 23

Why are fossil fuels not considered to be renewable?

A. Mining of fossil fuels is damaging to the surrounding environment and habitats.

B. Fossil fuels will eventually run out.

C. Fossil fuels are being used up faster than they can be formed.

D. Fossil fuels are not sustainable.

Question 24

Hydrogen is an energy-dense fuel source with many advantages, including its abundance on Earth in the form of water. Production of hydrogen gas from water requires energy. Hydrogen derived from which of the following forms of production would be classified as ‘green’ hydrogen?

A. from fossil fuels

B. from industrial processes

C. from fossil fuels with carbon capture

D. using renewable energy sources including wind

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Questions 25, 26 and 27 are based on the following information.

Ethanol can be produced industrially by reacting ethene with steam according to the equation

$CH_2CH_2(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g) \qquad \Delta H = -45 \text{ kJ mol}^{-1}$

Question 25

Ethanol production from ethene and steam requires temperatures of 300°C. The reason for this choice of temperature would be to:

A. increase the yield of ethanol.

B. increase the rate of ethanol production.

C. keep the system in gaseous form.

D. favour the forward reaction.

Question 26

Phosphoric acid is also used in this process, but is not included in the thermochemical equation. This is because:

A. phosphoric acid is a catalyst and does not take part in the reaction.

B. acids do not need to be shown in thermochemical equations.

C. phosphoric acid reacts with the ethanol to produce ethanoic acid.

D. all of the above.

Question 27

Tea tree oil has been used as an antiseptic for thousands of years by Indigenous Australians, but was only commercialised in 1920. The oil is separated from the biomass by heating to a vapour at 100°C and collected from a condenser at between 35°C and 55°C. The name given to this separation technique is:

A. distillation.

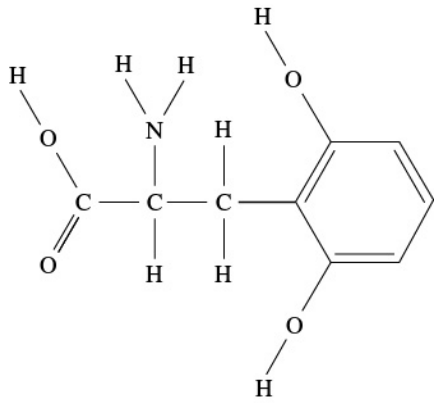
B. volumetric analysis.

C. vaporisation.

D. chromatography.

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 28



The molecule shown here is known as Dopa. It is used in the treatment of Parkinson's disease.

Which of the following functional groups is NOT present in Dopa?

A. hydroxyl

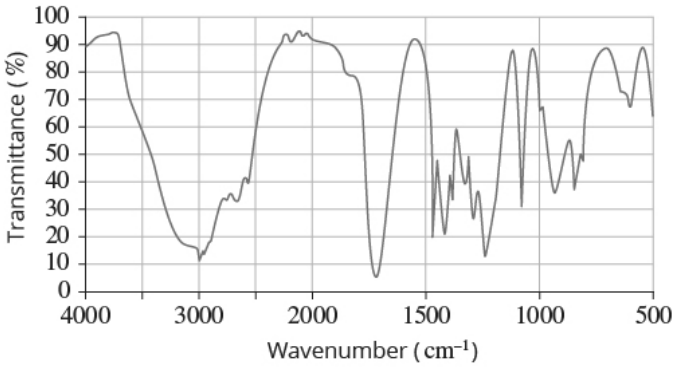
B. amine

C. carboxyl

D. amide

Question 29

Which functional group can be identified from bands in this IR spectrum?



A. carboxyl

B. hydroxyl

C. amine

D. amide

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 30

4.40 g of P_4O_6 and 3.00 g of I_2 are mixed and allowed to react according to the equation:

$$5P_4O_6(s) + 8I_2(s) \rightarrow 4P_2I_4(s) + 3P_4O_{10}(s)$$

Which reactant is in excess and by what mass?

A. P_4O_6 by 0.88 g

B. P_4O_6 by 1.4 g

C. P_4O_6 by 2.78 g

D. I_2 by 2.08 g

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Section B—Short-answer questions

11 questions

90 marks

Allow about 1 hour and 50 minutes for this section.

Instructions for Section B

Answer **all** questions in the spaces provided.

Give simplified answers to all numerical questions, with an appropriate number of significant figures; unsimplified answers will not be given full marks.

Show all working in your answers to numerical questions; no marks will be given for an incorrect answer unless it is accompanied by details of the working.

Ensure chemical equations are balanced and that the formulas for individual substances include an indication of state, for example, $H_2(g)$, NaCl(s).

Unless otherwise indicated, the diagrams in this exam are **not** drawn to scale.

Question 1 (7 marks)

Ethane is a large component in the natural gas that is used in Bunsen burners.

a. Write a thermochemical equation for the combustion of ethane. (2 marks)

.....

b. Calculate the energy released from the combustion of 5.00 g of ethane. (1 mark)

.....

.....

c. Calculate the net energy required to form 4.0 mol of ethane from carbon dioxide and water. (1 mark)

.....

.....

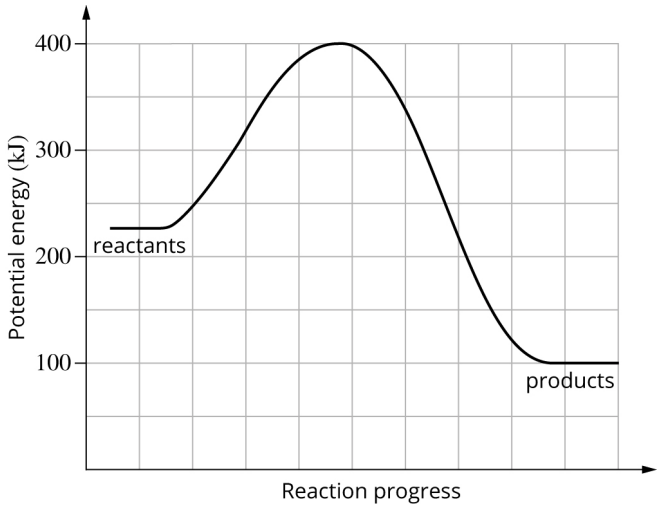
d. If a natural gas power station produces 4.97×10^{14} MJ of energy per day, but is only 53% efficient, how much natural gas, in kg, must be burnt to supply this energy? (2 marks)

.....

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

The energy profile diagram for the thermochemical equation for the complete combustion of ethane is shown.



e. On the graph shown here, draw a dotted line to show how the energy profile diagram is changed by the use of a catalyst. (1 mark)

Question 2 (13 marks)

Nitrogen monoxide produced in the combustion engines of cars can react with oxygen in the air to form the brown pollutant nitrogen dioxide. The equation is:

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

1.60 mol of nitrogen monoxide and 1.60 mol of oxygen gas are added to an empty 1.00 L reaction vessel. At equilibrium, 0.48 mol of NO_2 has formed.

a. Calculate the value of the equilibrium constant, *K*, of the mixture. (3 marks)

……

……

……

……

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

b. Two different adjustments are made to equilibrium mixtures of the above gases. Use the table provided to indicate the impact of each change. (6 marks)

	Impact upon *K*	**Impact on the amount of CO**	**Impact on the concentration of CO**
Decrease in temperature			
Decrease in volume			

The equation for the reaction between hydrogen and nitrogen monoxide gases is:

$$2H_2(g) + 2NO(g) \rightleftharpoons 2H_2O(g) + N_2(g) \qquad K = 12.4 \text{ at } 40°C \qquad \Delta H = -132 \text{ kJ mol}^{-1}$$

c. Calculate the values of K and ΔH for the following reactions at 40°C:

i. $2H_2O(g) + N_2(g) \rightleftharpoons 2H_2(g) + 2NO(g)$ (1 mark)

.....

.....

ii. $H_2O(g) + \frac{1}{2}N_2(g) \rightleftharpoons H_2(g) + NO(g)$ (1 mark)

.....

.....

d. The numerical value of K for a reaction involving the above gases is 3.52 at 40°C. What reaction will have a K with this value? (2 marks)

.....

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 3 (7 marks)

Lithium is a metal that is often used in batteries. To investigate the potential dangers of the lithium in the batteries coming into contact with water, a calorimeter was used.

a. To calibrate the calorimeter, 8.20 V and 750 mA was run through the calorimeter's electric heating coil, operating for 10.0 minutes. This resulted in a temperature increase of 5.88°C. Calculate the calibration factor. (2 marks)

.....

.....

.....

.....

b. 0.283 g of solid Li metal is added to the water in the calorimeter, with the reaction producing a temperature increase from 18.45°C to 32.38°C. Calculate the enthalpy change, in kJ mol^{-1}, for the equation:

$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$ (3 marks)

.....

.....

.....

.....

c. Green chemistry principles relate to sustainability, and include the use of renewable feedstocks and designing safer chemicals. Discuss the continued use of lithium in batteries in terms of these principles. (2 marks)

.....

.....

.....

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 4 (8 marks)

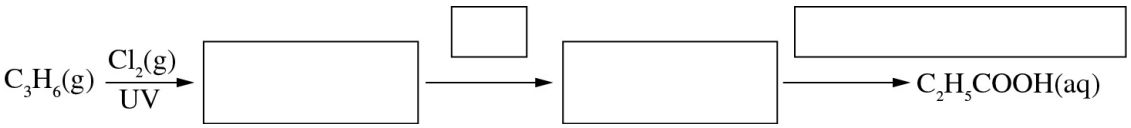
Propanoic acid and hex-1-ene are both clear liquids. Chemical tests can be performed to distinguish between these two substances.

a. Use the table provided to:

- explain a chemical test that could be conducted
- state expected conclusions
- write a relevant chemical equation for any reaction occurring. (4 marks)

Test	Experiment conducted	Likely result	Chemical equation
1			
2			

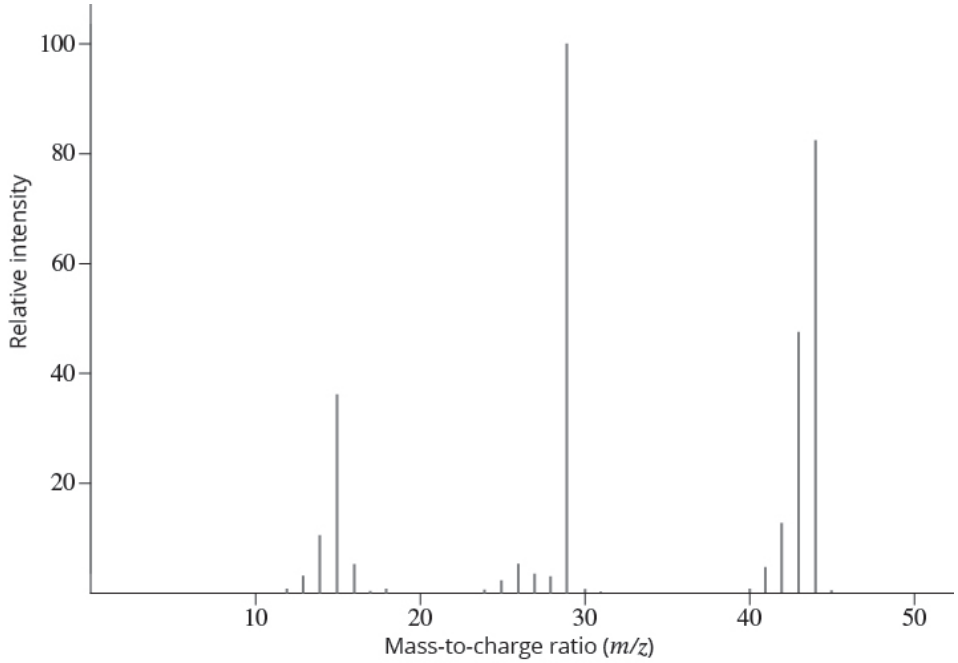
b. Complete the boxes below to show a synthesis pathway for the production of propanoic acid from propane. Use a semi-structural formula for the products. (4 marks)



Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 5 (7 marks)

a. The mass spectrum of a molecule containing carbon, hydrogen and oxygen is shown below.



i. Suggest a molecular formula for this molecule. (1 mark)

.....

.....

ii. Suggest possible fragments that might cause the peaks at: (2 marks)

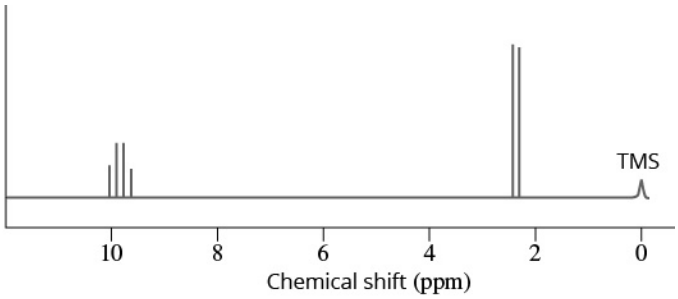
15: .....

29: .....

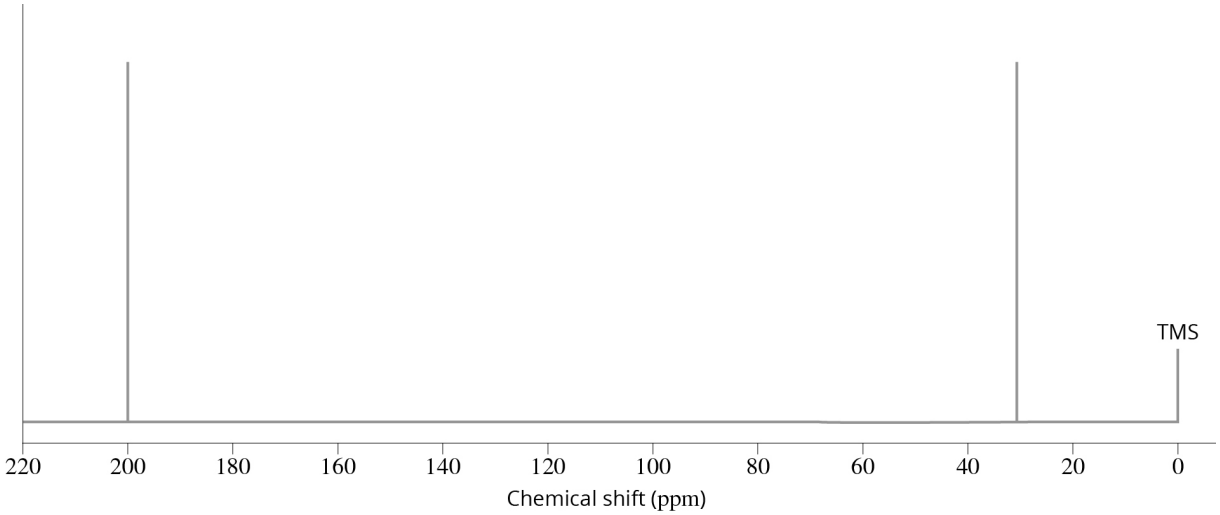
Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

The proton (^{1}H) NMR and the carbon-13 NMR of the molecule are both shown below.

Proton NMR



Carbon-13 NMR



b. State one piece of information from each spectrum that you used to help you identify the unknown compound.

i. ^{1}H NMR (1 mark)

.....

.....

ii. ^{13}C NMR (1 mark)

.....

.....

c. Name and draw the skeletal structure of the unknown compound. (2 marks)

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 6 (6 marks)

Rhubarb stalks can be bought from fruit and vegetable shops and incorporated into desserts or pies. The stalks have a tart taste due to the oxalic acid they contain. The chemical formula of oxalic acid is $C_2H_2O_4$.

Oxalic acid can react with purple solutions of potassium permanganate, turning the purple solutions colourless. Adding solid pieces of rhubarb stalk to potassium permanganate solutions will gradually cause the solutions to become colourless. The equation for the reaction occurring is:

$$2MnO_4^-(aq) + 5C_2H_2O_4(aq) + 26H^+(aq) \rightarrow 2Mn^{2+}(aq) + 10CO_2(g) + 18H_2O(l)$$

purple colourless

The time taken for the purple colour to disappear once the rhubarb is added provides an indication of the rate of this reaction. A student investigating this reaction sums up her results in the table below. The same volume and concentration of potassium permanganate solution is used each time.

Experiment	Temperature (°C)	Rhubarb preparation	Time for solution to become colourless (s)
1	20	10 cm stalk added as one piece	85
2	20	10 cm stalk sliced into four pieces	78
3	20	10 cm stalk crushed	52
4	40	10 cm stalk added as one piece	22

a. Use collision theory to explain the trend in times recorded in experiments 1, 2 and 3. (1 mark)

.....

.....

b. Use collision theory to explain the difference in times obtained between experiment 1 and experiment 4. (2 marks)

.....

.....

.....

c. In this experiment, what is the dependant variable? (1 mark)

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

d. Suggest two things that could be done to the method to increase the validity of this experiment. (2 marks)

.....

.....

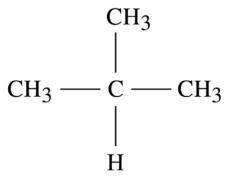
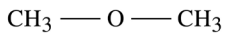
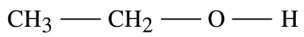
.....

Question 7 (8 marks)

A student was asked to separate three substances currently mixed together in a solution bottle.

The student decided to use fractional distillation, separating the substances by boiling point.

a. Explain which of the following compounds has the highest boiling point. (3 marks)

propane	
methoxymethane	
ethanol	

.....

.....

.....

.....

.....

.....

b. Which solution would the student collect in the first fraction? (1 mark)

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

To confirm their identification of the three fractions, the student decided to use high-performance liquid chromatography (HPLC).

c. Why do compounds have different retention times in HPLC? (2 marks)

.....

.....

.....

.....

.....

d. If a polar solvent was used as the mobile phase, which of the three substances would have the longest retention time? Explain your response. (1 mark)

.....

.....

The student was then asked to analyse the purity of a propane gas sample. To do this, they first ran a sample of hydrocarbons known to contain pure propane, Figure 1. The unknown sample, Figure 2, was then run in identical conditions.

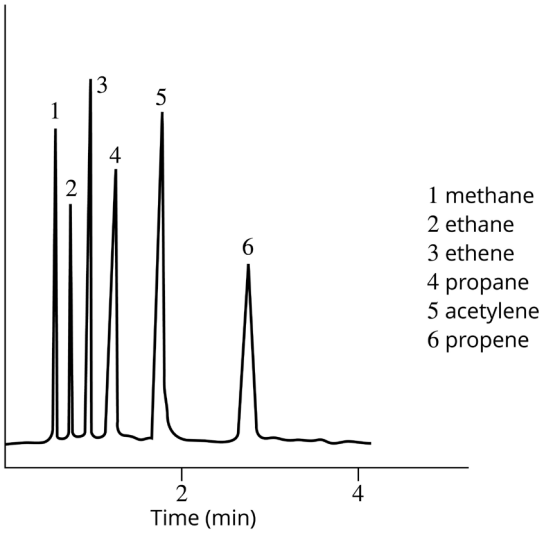


Figure 1

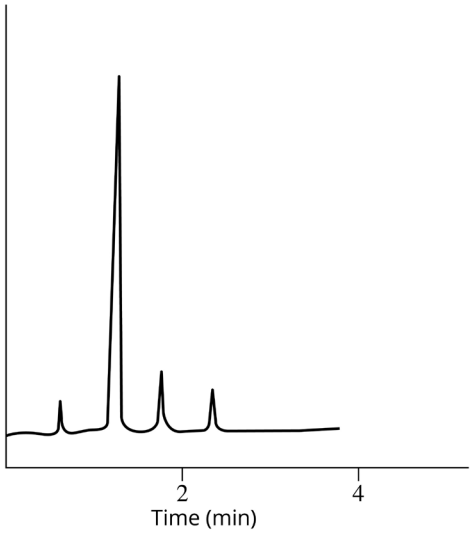


Figure 2

e. Use the gas chromatogram of the unknown sample, Figure 2, to evaluate its purity. (1 mark)

.....

.....

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

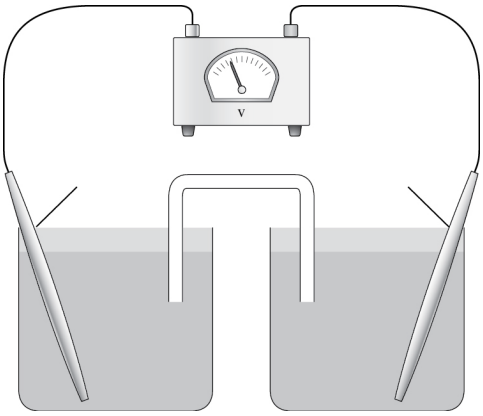
Question 8 (9 marks)

An electrochemical cell is made from Sn^{2+}/Sn and Fe^{2+}/Fe half-cells under standard conditions.

a. Determine the overall cell reaction. (1 mark)

.....

b. Identify the anode, cathode and direction of electron flow by labelling the diagram below. (3 marks)



c. Explain the function of the salt bridge in an electrochemical cell. (2 marks)

.....

.....

.....

.....

d. Determine the standard electrical potential this cell would be predicted to produce. (1 mark)

.....

e. When the cell is running, it only produces 0.20 V. Identify two limitations associated with the use of standard electrode potentials. (2 marks)

.....

.....

.....

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

Question 9 (4 marks)

When beryllium is added to an acidified solution of potassium permanganate ($KMnO_4$), a reaction occurs that produces beryllium ions (Be^{2+}) and manganese ions (Mn^{2+}).

a. Construct the oxidation and the reduction half-reactions for the reaction. (2 marks)

Oxidation: .....

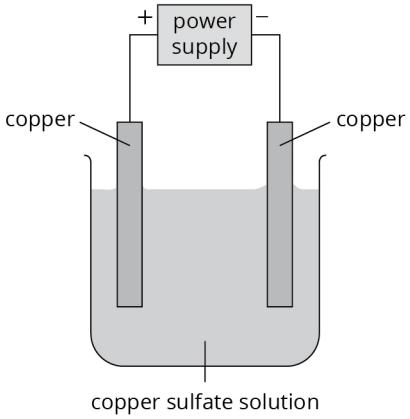
Reduction: .....

b. Construct the balanced chemical equation under acidic conditions for the reaction. (2 marks)

.....

Question 10 (8 marks)

The electrolysis of a copper(II) sulfate solution was set up using inert electrodes.



a. Predict the initial products at the anode and the cathode. (2 marks)

Anode: .....

Cathode: .....

b. Assuming the cell is 100% efficient, calculate how long the cell will need to operate for 10.0 g of Cu to be produced if the cell used 2.40 A. (4 marks)

.....

.....

.....

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

c. Unfortunately, the cell was only 80% efficient. Calculate the length of extra time the cell would actually need to run to produce 10.0 g of Cu. (2 marks)

.....

.....

.....

.....

Question 11 (13 marks)

a. Bioethanol can be considered to be a carbon-neutral fuel. Explain this viewpoint, incorporating chemical equations for photosynthesis and fermentation in your answer. (3 marks)

.....

.....

.....

.....

b. Researchers have also found a way to use artificial photosynthesis to provide alternative electricity and liquid fuel options. Write the two redox half equations used in the proton reduction catalyst system. (2 marks)

.....

.....

c. Explain how artificial photosynthesis can be said to take advantage of a renewable feedstock. (1 mark)

.....

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8

d. Fats and oils are another fuel source that can be used to help solve the need for liquid fuels. Draw the general structure of a lipid. (2 marks)

e. Circle and name the major functional group found in lipids. (1 mark)

.....

f. To be used as a fuel in transport, fuels must be able to flow. Unsaturated fats such as olive oil are liquid at room temperature while saturated fats such as butter are solid. Explain this difference in melting points by referring to the structural differences between these fats. (2 marks)

.....

.....

.....

.....

g. If 17.50 g of a fatty acid with a molar mass of 280.0 g mol^{-1} becomes fully saturated with exactly 3.10 L of I_2 gas at SLC, determine the degree of unsaturation of the fatty acid. (2 marks)

.....

.....

.....

.....

Copyright © Pearson Australia 2023 (a division of Pearson Australia Group Pty Ltd) ISBN 978 0 6557 0009 8